

Quantum Mechanics : : Homework 08

April 1, 2025

Ahmed Saad Sabit, Rice University

Extension granted.

Problem 01

Used in Integral Calculator:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar}$$

For $n = 1$ eigenstate wavefunction keeping a common factor between two wavefunctions

$$\psi_1(x) = x \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \sqrt{\frac{2m\omega}{\hbar}} e^{-m\omega x^2/2\hbar}$$

(a)

Without $C\delta(x_1 - x_2)$ repulsion, the two particles could happily have the same wave functions $\psi(x)\sigma_{\text{spin}}$. Without the perturbation

$$\hat{H} |\psi_1, \psi_2\rangle = \hat{H}_1 |\psi_1 \psi_2\rangle + \hat{H}_2 |\psi_1 \psi_2\rangle \implies \varepsilon |\psi_1 \psi_2\rangle$$

$$E_0^{[0]} = 2 \left(\frac{\hbar\omega}{2}\right) = \hbar\omega$$

With the perturbation

$$\hat{H} |\psi_1, \psi_2\rangle = \hat{H}_1 |\psi_1 \psi_2\rangle + \hat{H}_2 |\psi_1 \psi_2\rangle + C\delta(\hat{x}_1 - \hat{x}_2) |\psi_1 \psi_2\rangle \implies E |\psi_1 \psi_2\rangle$$

The energy level perturbation of first order

$$\Delta = \langle \psi_1 \psi_2 | C\delta(\hat{x}_1 - \hat{x}_2) | \psi_1 \psi_2 \rangle = C \int \Psi(x_1, x_1)^* \Psi(x_1, x_1) dx_1$$

Unperturbed lowest energy state (using integral calculator)

$$C \int_{-\infty}^{\infty} dx |\Psi(x)|^4 = C \sqrt{\frac{m\omega}{2\pi\hbar}}$$

$$E_0^{[1]} = \hbar\omega + C \sqrt{\frac{m\omega}{2\pi\hbar}}$$

(b)

Energy without perturbation

$$E = \frac{1}{2}\hbar\omega + \frac{3}{2}\hbar\omega = 2\hbar\omega$$

Symmetric orbital is given by

$$\psi(x_1, x_2)_s = \frac{1}{\sqrt{2}} (\psi_0(x_1)\psi_1(x_2) + \psi_0(x_2)\psi_1(x_1))$$

Anti-Symmetric orbital is given by

$$\psi(x_1, x_2)_A = \frac{1}{\sqrt{2}} (\psi_0(x_1)\psi_1(x_2) - \psi_0(x_2)\psi_1(x_1))$$

It's easy compute the energy for anti-symmetric wavefunction because $\psi(x_1, x_2)_A = 0$ for $x_1 = x_2$ giving $E_A^{(1)} = 0$. The orbital wavefunction is anti-symmetric for spin triplet state.

For symmetric orbital wavefunction

$$E_S^{(1)} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx_1 dx_2 \psi_S^*(x_1, x_2) C \delta(x_1 - x_2) \psi_S(x_1, x_2) = \int_{-\infty}^{\infty} dx 2C \psi_0^2(x) \psi_1^2(x) = C \sqrt{\frac{m\omega}{2\pi\hbar}}$$

The 4 degeneracy of the energy has been converted to 3 degeneracy in Anti-symmetric orbital wave function (triplet spin state) and there's only 1 state in single case with a different energy given (both) by

$$\boxed{E_S^{(1)} = C \sqrt{\frac{m\omega}{2\pi\hbar}} \quad \text{and} \quad E_A^{(1)} = 0}$$

Problem 02

(a)

Plugging in $K = 1$ hence solving the equation for expectation value is with $Z_e = 2$

$$2 * 2 \frac{1+3+1}{2^3} (13.6 \text{ eV}) = 34.0 \text{ eV}$$

(b)

$$\begin{aligned} \langle H \rangle &= \langle \psi_{100} \psi_{100} | \frac{p^2}{2m} + \frac{P^2}{2M} - \frac{2e^2}{R} - \frac{2e^2}{r} + \frac{e^2}{|\vec{r} - \vec{R}|} | \psi_{100} \psi_{100} \rangle \\ &= E_{\text{electron ground}} + E_{\text{muonic ground}} + E_{\text{interaction}} \end{aligned}$$

Using 11.66 and 11.72 in Townsend we can find

$$\begin{aligned} \langle nlm | \hat{p}^2 / 2\mu | nlm \rangle &= -E_n^{(0)} = \frac{\mu c^2 Z^2 \alpha^2}{2n^2} \implies E_n^{(0)} \approx -\frac{m_e c^2 Z^2 \alpha^2}{2n^2} \\ -\frac{1}{2} \langle nlm | \frac{Ze^2}{\hat{r}} | nlm \rangle &= E_n^{(0)} \implies \langle nlm | \frac{Ze^2}{\hat{r}} | nlm \rangle = -2E_n^{(0)} \approx \frac{m_e c^2 Z^2 \alpha^2}{n^2} \end{aligned}$$

Invoke R_∞ then

$$E_n^{(0)} = -\frac{Z^2 R_\infty}{n^2} = -\frac{1}{n^2} \left(\frac{m_e c^2 Z^2 \alpha^2}{2} \right)$$

$$-2E_n^{(0)} = \frac{2ZR_\infty}{n^2}$$

So for electron $E_n^{(0)}$

Total kinetic energy term for electron and muon

$$E_k^{(0)} = R_\infty \left(Z_e^2 + Z_\mu^2 \frac{M}{m} \right)$$

Total potential energy

$$E_p^{(0)} = -4R_\infty \left(Z_e + Z_\mu \frac{M}{m} \right)$$

Total Energy

$$E = R_\infty \left(Z_e^2 + Z_\mu^2 \frac{M}{m} \right) - 4R_\infty \left(Z_e + Z_\mu \frac{M}{m} \right) + 2Z_e \frac{(1 + 3K + K^2)}{(1 + K)^3} R_\infty$$

(c)

Because of the following fact

$$\left| \frac{d}{dK} \frac{1 + 3K + K^2}{(1 + K)^3} \right|_{K=0} = 0$$

We can make the approximation

$$0 = \frac{\partial E}{\partial Z_e} = 2R_\infty Z_e - 4R_\infty + 2R_\infty \implies Z_e = 1$$

$$0 = \frac{\partial E}{\partial Z_\mu} = 2Z_\mu R_\infty \frac{M}{m} - 4R_\infty \frac{M}{m} \implies Z_\mu = 2$$

$$K = \frac{1}{2} \frac{m}{207m} = \frac{1}{414} \implies \frac{1 + 3K + K^2}{(1 + K)^3} \approx 1$$

$$E_{\text{ground state}} = -829R_\infty$$

(d)

Given the higher mass of muon than electron, we should expect the muon to be much closer to the nucleus than the electron (in a somewhat semi-quasi-classical preview). The effective charge hence shifts and the electron witnesses $Z_e = 1$ and $Z_\mu = 2$

Problem 03

(a)

$$1s^2 2s^2 2p^6 3s^2 3p^3$$

We can count 3 unpaired valence electron which means total spin is $\frac{3}{2}$ giving us $2s + 1 = 4$.

$L_{\text{tot}} = |-1 + 0 + 1| = 0$. The three valence electrons are at different states so $J = |L - S| = \frac{3}{2}$. The shell being half full we know $^4S_{3/2}$

(b)

$$1s^2 2s^2 2p^6 3s^2 3p^4$$

$$S_{\text{tot}} = (2 \text{ in pair, } 2 \text{ without}) = 1 \implies 2S + 1 = 3$$

$$L_{\text{tot}} = |-1 + 0 + 1 \pm 1| = 1 \text{ which is } P$$

$$J = L + S = 2 \implies {}^3P_2$$

(c)

$$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$$

$$S_{\text{tot}} = (3 \text{ without pairing}) = \frac{3}{2} \implies 2S + 1 = 4$$

$$L_{\text{tot}} = |-2 - 1 + 0 + 1 + 2 - 2 - 1 + 0| = 3 \text{ which is } F$$

$$J = 3 \text{ unpaired hence } = |J - S| = \frac{3}{2} \implies {}^4F_{3/2}$$

(d)

$$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$$

$$S_{\text{tot}} = (2 \text{ without pair}) = 1 \implies 2S + 1 = 3$$

$$L_{\text{tot}} = |-2 - 1 + 0 + 1 + 2 - 2 - 1 + 0| = 3 \text{ which is } F$$

$$J = L + S = 4 \implies {}^3F_4$$